

Milestone-B: Automation Execution Integrity

Verification of a separate TestNG execution lane

1 Document details

Field	Recorded value
Prepared by	Darvin Anorma Mangampo
Assessment period	11-12 September 2026 (UTC+08:00)
Repository	selenium-saucedemo-remediation
Document date	12 September 2026
Technical commit	882a08a7fec606855baf556ddffd25fd6105dda5

2 Purpose

Milestone-B began at the published Milestone-A commit. I added a separate maintainedTest lane for one non-browser TestNG probe, then checked how Gradle reported the result requested from the command line.

The first run used integrityProbe=fail. I kept its console output and XML result before running the same source with integrityProbe=pass. After both runs, I compared the invocation counts and checked the saved hashes for the source files and tracked workbook. I also checked the repository state and committed paths at closure.

3 Repository origin and change boundary

The application source and original test setup are inherited, so I do not claim them as my work. I began this milestone from the completed Milestone-A commit shown below. My Milestone-B technical implementation consists of the Gradle definition for the separate lane and the probe source.

Boundary	Recorded position
Starting commit	d48d41dbc53f3e675089f537bc2c45bc9980deea
Inherited baseline	inherited-baseline-2026-09-05 at 4929e146a856d3b9ca2f04f5fda2eb014f599aa4
Milestone-B technical implementation	build.gradle and src/maintainedTest/java/integrity/ExecutionIntegrityProbe.java
Execution boundary	Separate non-browser TestNG probe only

4 Activity record

Activity	Work performed	Recorded result
M-B-A1	Checked the repository against Milestone-A. The workbook and existing toolchain were then confirmed.	The worktree was clean at the Milestone-A commit, and the workbook hash matched. Java 14.0.2 and Gradle 6.6.1 were available.
M-B-A2	Added a separate source set and task. TestNG was declared directly for the lane, with one probe class.	The source compiled with exit code 0. Both maintained classpaths contained TestNG 7.3.0, while inherited test compilation did not run.
M-B-A3	Ran the integrity probe with integrityProbe=fail and retained its console and XML records. Then ran the same unchanged probe with integrityProbe=pass.	The run made with integrityProbe=fail returned exit code 1. The following run, made with integrityProbe=pass, returned exit code 0. Each retained XML result recorded one invocation, so no retry was observed in these two runs. The source file hashes and workbook hash remained unchanged.
M-B-A4	I reviewed the two-file technical diff and signed the commit. After the push, I checked the remote state.	GitHub marked commit 882a08a7... as Verified. It matched origin/master, and the worktree was clean when the activity closed.

5 Maintained lane implementation

The `maintainedTest` source set reads Java from `src/maintainedTest/java` and declares TestNG 7.3.0 directly. Its Gradle task uses the source set's compiled classes and runtime classpath. The maintained configuration does not extend an inherited dependency bucket.

The task selects only `ExecutionIntegrityProbe`. It keeps `ignoreFailures` at `false`, so a test failure reaches Gradle. No inherited XML suite is configured for this task. The inherited retry transformer and listeners are outside its classpath. Hooks and workbook helpers are outside the lane too. It contains no browser code. The `integrityProbe` property supplies the requested outcome. Without that property, the probe uses `pass`.

Item	Recorded setup
Source set	<code>maintainedTest</code>
Test source	<code>src/maintainedTest/java/integrity/ExecutionIntegrityProbe.java</code>
Dependency	<code>maintainedTestImplementation 'org.testng:testng:7.3.0'</code>
Task	<code>maintainedTest</code> of type <code>Test</code> , using TestNG with <code>ignoreFailures = false</code>
Task input	Command-line property <code>integrityProbe</code> ; default value <code>pass</code>
Task graph	<code>compileMaintainedTestJava</code> , <code>processMaintainedTestResources</code> , <code>maintainedTestClasses</code> , <code>maintainedTest</code>

Both maintained classpath reports contained TestNG 7.3.0. Neither report matched the blocked inherited dependencies. The dry-run graph contained only the four maintained tasks. `compileTestJava` and `test` were absent. The cucumber task did not appear either. Before the probe ran, `maintainedTestClasses` finished with exit code 0.

6 Failure and pass verification

Check	Failure request	Passing request	Comparison
Process result	Exit 1; build and task failed	Exit 0; build and task passed	The process result followed the TestNG result.
Test result	1 test; 1 failure	1 test; 1 pass	Only <code>ExecutionIntegrityProbe</code> ran.
Invocation count	1	1	No retry was observed in either retained XML result.
Execution freshness	Not cached	Not cached	Both runs used <code>rerun-tasks</code> and <code>no-build-cache</code> .
Source state	Starting hashes retained	Same hashes retained	The source was unchanged between runs.

7 Repository closure

Check	Recorded result
Technical commit	<code>882a08a7fec606855baf556ddffd25fd6105dda5</code>
Parent commit	<code>d48d41dbc53f3e675089f537bc2c45bc9980deea</code>
Committed paths	<code>2: build.gradle</code> and <code>src/maintainedTest/java/integrity/ExecutionIntegrityProbe.java</code>
Workbook SHA-256	<code>9ECECF817507E4762A560DD1BFDA6C563B79648114B4121C1C54906CFDA07B9C</code>
Final state	The worktree was clean after local HEAD matched origin/master. The technical commit contained no generated path.

8 Limits

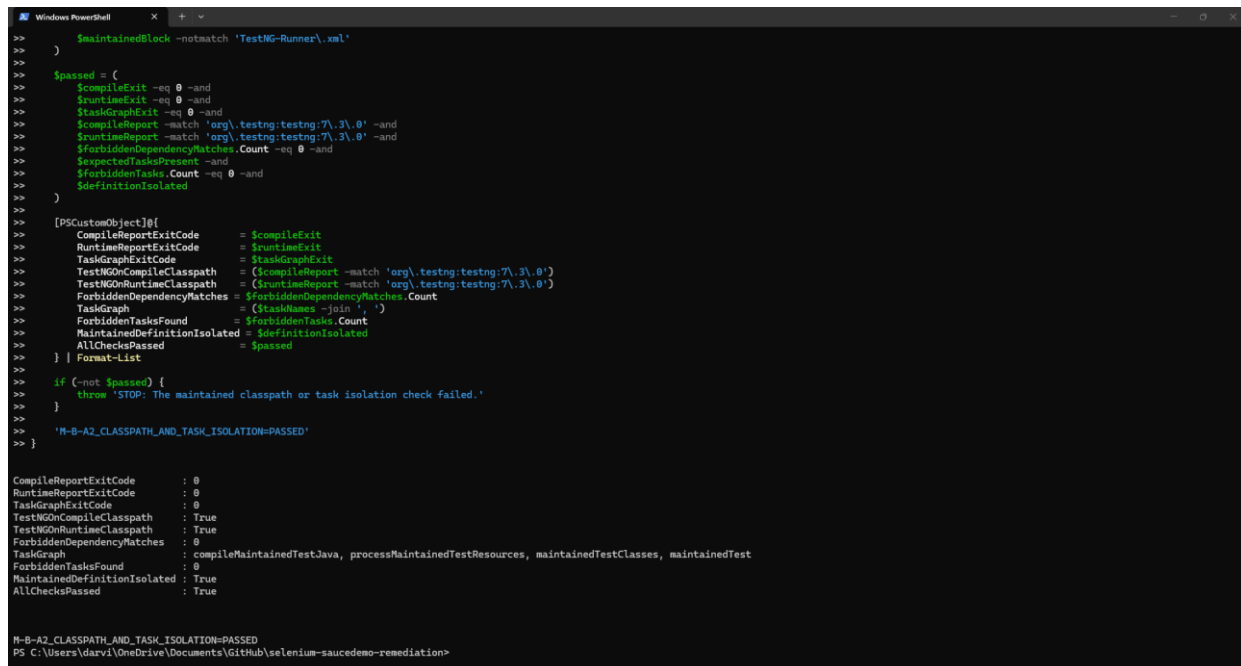
These results apply to the separate maintained non-browser lane. They do not show whether the inherited test source compiles. Repair of the legacy XML suites was not assessed here. Selenium and browser readiness were also outside this milestone.

No application function was tested in Milestone-B. The intentional failure came from the integrity probe, so it is not an application defect. The retry statement is limited to the two retained runs. Each result recorded one invocation, with no retry observed.

9 Supporting screenshots

Five screenshots are included. Figure 1 records the classpath and task check. Figures 2 and 3 show the requested runs. GitHub verification appears in Figure 4, followed by the final repository check.

Maintained classpath and task isolation



```

>> $maintainedBlock -notmatch 'TestNG-Runner.xml'
>> }
>> $passed = (
>> $compileExit -eq 0 -and
>> $runtimeExit -eq 0 -and
>> $taskGraphExit -eq 0 -and
>> $compileReport -match 'org.testng.testng:7.3.0' -and
>> $runtimeReport -match 'org.testng.testng:7.3.0' -and
>> $forbiddenDependencyMatches.Count -eq 0 -and
>> $expectedTasksPresent -and
>> $forbiddenTasks.Count -eq 0 -and
>> $definitionIsolated
>> )
>> [PSCustomObject]@{
>>   CompileReportExitCode = $compileExit
>>   RuntimeReportExitCode = $runtimeExit
>>   TaskGraphExitCode = $taskGraphExit
>>   TestNGOnCompileClasspath = ($compileReport -match 'org.testng.testng:7.3.0')
>>   TestNGOnRuntimeClasspath = ($runtimeReport -match 'org.testng.testng:7.3.0')
>>   ForbiddenDependencyMatches = $forbiddenDependencyMatches.Count
>>   TaskGraph = ($taskNames -join ', ')
>>   ForbiddenTasksFound = $forbiddenTasks.Count
>>   MaintainedDefinitionIsolated = $definitionIsolated
>>   AllChecksPassed = $passed
>> } | Format-List
>>
>> if (-not $passed) {
>>   throw 'STOP: The maintained classpath or task isolation check failed.'
>> }
>>
>> 'M-B-A2_CLASSPATH_AND_TASK_ISOLATION=PASSED'
>> }

CompileReportExitCode : 0
RuntimeReportExitCode : 0
TaskGraphExitCode : 0
TestNGOnCompileClasspath : True
TestNGOnRuntimeClasspath : True
ForbiddenDependencyMatches : 0
TaskGraph : compileMaintainedTest.java, processMaintainedTestResources, maintainedTestClasses, maintainedTest
ForbiddenTasksFound : 0
MaintainedDefinitionIsolated : True
AllChecksPassed : True

M-B-A2_CLASSPATH_AND_TASK_ISOLATION=PASSED
PS C:\Users\darvi\OneDrive\Documents\GitHub\selenium-saucedemo-remediation>

```

Figure 1. Both maintained classpaths contain TestNG 7.3.0. The task graph has four maintained tasks and no blocked dependency match.

Failure and passing results

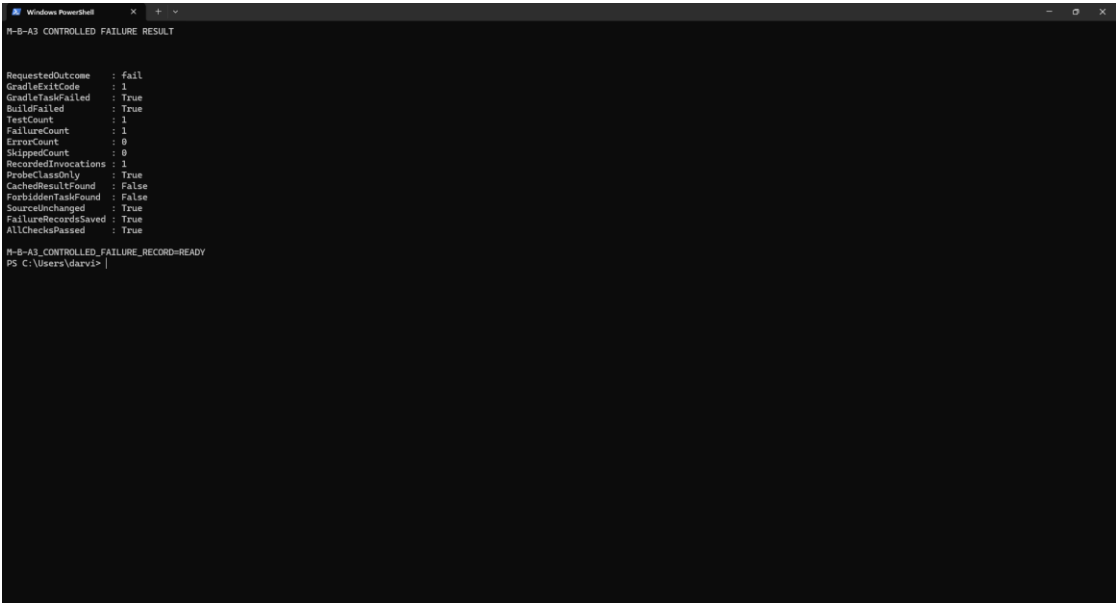


Figure 2. One test failed as requested. Gradle returned exit code 1 and marked the task as failed.

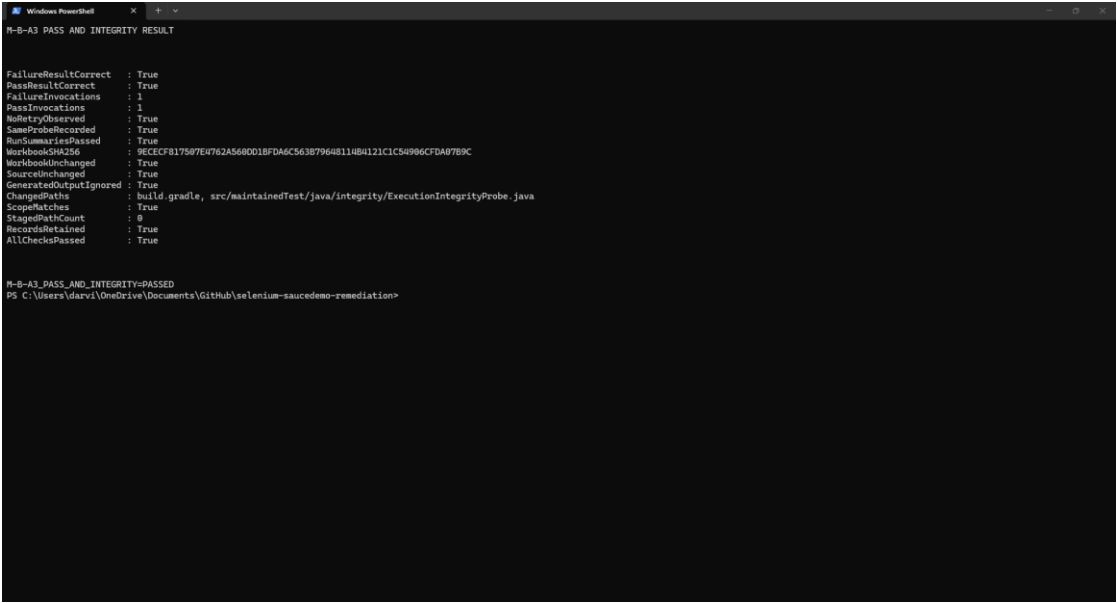


Figure 3. The comparison retains one invocation from each run and shows no retry. Workbook and source hashes remain unchanged.

Commit and final state

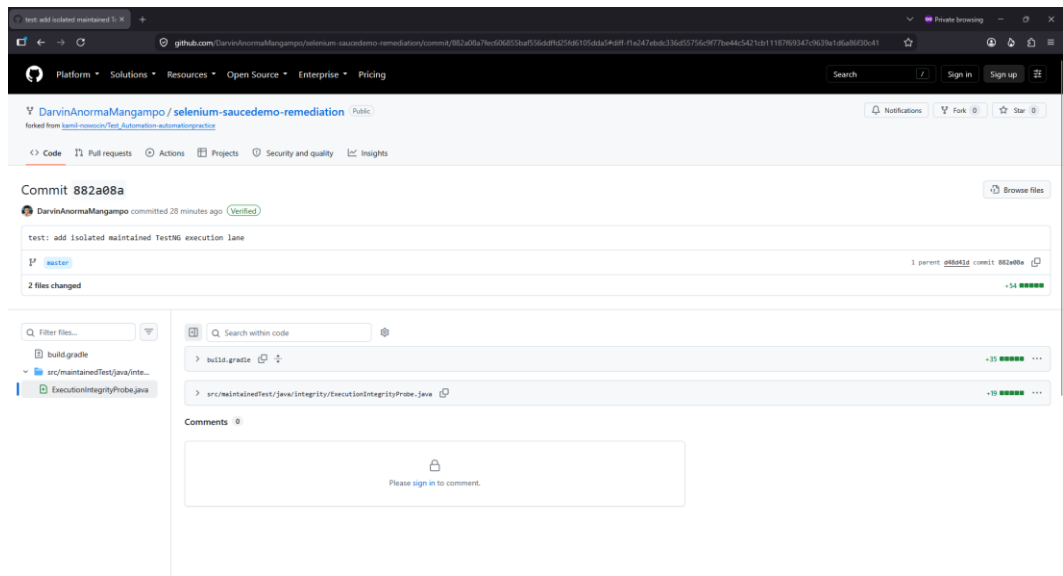


Figure 4. Signed-out GitHub view of commit 882a08a7 with the Verified badge and two changed files.

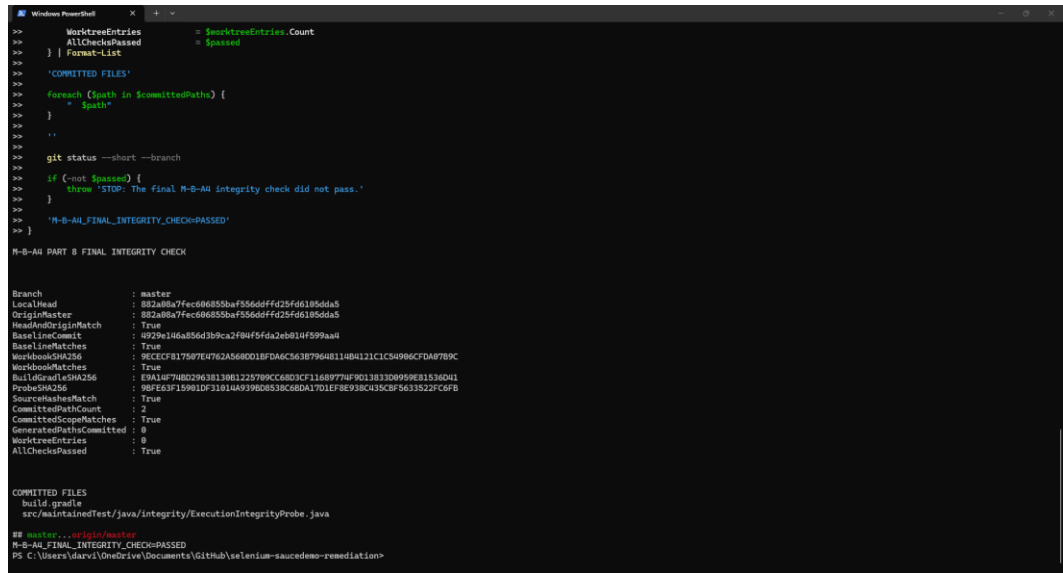


Figure 5. Final repository check after the push. Local HEAD matches origin/master and the worktree is clean.

10 Conclusion

The separate maintained TestNG lane returned both requested results through the Gradle process.

One run intentionally failed and returned exit code 1. The same source then passed with exit code 0. Both XML results contain one invocation, with no retry observed.

The tracked workbook kept the same SHA-256 value. Generated classes and reports remained under build output that Git ignores. Technical commit 882a08a7fec606855baf556ddffd25fd6105dda5 contains only build.gradle and the probe source. When technical closure was recorded, local HEAD matched origin/master.

Repository references

- [Milestone-B technical commit](#)
- [build.gradle at the Milestone-B commit](#)
- [ExecutionIntegrityProbe.java at the Milestone-B commit](#)
- [Inherited baseline tag](#)